INTERNATIONAL MONETARY FUND

KINGDOM OF LESOTHO

DEBT SUSTAINABILITY ANALYSIS UPDATE

May 19, 2014

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This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from September 2013. It incorporates macroeconomic projections consistent with the government budget for 2014/15, which was recently approved by parliament, and employs the 5 percent discount rate that is now standard for present value calculations. Results indicate that Lesotho remains at moderate risk of external debt distress.[1] *However, these results are sensitive to the assumptions concerning the financing of the hydropower plant under the second phase of the Lesotho Highlands Water Project and projections of government's annual financing needs. Given the low level of domestic public debt, the debt indicator for total public debt remains well below its relevant benchmark.*

[1] The low-income country debt sustainability framework (LIC DSF) recognizes that better policies and institutions allow countries to manage higher levels of debt, and thus the threshold levels are policy-dependent. Lesotho's policies and institutions, as measured by the World Bank's Country Policy and Institutional Assessment (CPIA), place it as a "medium performer", with an average rating of 3.45 during 2010-12. The relevant indicative thresholds for this category are: 40 percent for the present value (PV) of debt-to-GDP ratio, 150 percent for the PV of debt-to-exports ratio, 250 percent for the PV of debt-to-revenue ratio, 20 percent for the debt service-to-exports ratio, and 20 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

UNDERLYING ASSUMPTIONS

1. Based on recent revisions to historical national accounts data and revised assumptions concerning the timeline for construction of the water and hydropower components of the second phase of the Lesotho Highlands Water Project (LHWP-2), real GDP growth is projected to dip in 2014 before picking up over the medium term. On average, growth rates are slightly higher in 2015–17, compared with projections for the previous DSA. Over the long term, the average annual growth rate was revised downwards from 4.8 percent to 4.5 percent, mainly reflecting a lower share of capital spending in the government's budget. For imports, the underlying long-term trend anticipates a slight import substitution effect with growth, while over the medium term increases in imports would outpace GDP growth because of imported inputs for LHWP-2 construction. Export growth over the medium term is largely determined by expected developments in the mining and manufacturing sectors. Over the long term exports would receive a boost from the completion of LHWP-2.

Macroeconomic Assumptions, 2012-2033

	2012	2013	2014	2015	2016	2017	2018	2019-33 Average
	Act.	Est.			Projections			
Real GDP Growth (percent)	6.7	5.4	4.0	4.9	5.0	5.5	5.5	4.7
GDP deflator in U.S. dollar terms (change, in percent)	-11.6	-4.9	2.9	3.2	1.8	1.5	1.2	1.9
Effective interest rate (percent)	1.2	1.3	1.8	2.3	2.7	2.4	2.4	1.7
Growth of exports of goods and services (U.S. dollar terms, percent	-9.2	-6.9	7.8	4.5	12.1	9.6	6.5	7.0
Growth of imports of goods and services (U.S. dollar terms, percent	-2.9	-5.9	4.9	10.8	6.3	4.9	6.8	6.2
Grant element of new public sector borrowing (percent)	...	0.0	40.4	34.0	33.7	28.9	26.1	37.7
Government revenue (excluding grants, percent of GDP)	57.5	53.6	56.2	54.4	51.5	50.8	50.2	48.3
Government expenditure (percent of GDP)								
Primary (noninterest) expenditure	60.3	59.5	61.5	58.4	57.1	57.1	56.7	56.1
Interest payments								
Aid flows (millions of U.S. dollars)	942	928	855	824	800	780	758	520
of which:								
Grants	199	124	105	102	104	113	121	207
Concessioanl loans	744	804	750	721	695	668	637	313
Grant equivalent financing (percent of GDP)	...	5.3	6.5	5.9	5.5	5.6	5.7	5.6
Grant equivalent financing (percent of external financing)	...	100.0	66.0	59.1	59.8	53.8	50.0	62.2

2. On the fiscal side, the government revenue (excluding grants) was revised upward on average by 2 percent of GDP a year over the long run, largely reflecting higher revenue projections from the Southern African Customs Union (SACU) – compared with projections in the previous DSA. Primary (noninterest) expenditure projections have also been increased to reflect the latest projections in the authorities' medium-term fiscal framework, resulting in an average primary deficit of 4.3 percent of GDP per annum, which is assumed to be financed mostly through external concessional borrowing.

3. The assumptions regarding the hydropower component of LHWP-2 remain broadly the same as those in the previous DSA with respect to costs ($500 million) and financing terms and conditions (80 percent commercial, 20 percent grants). Only the timeline has been pushed back to 2019–26, in light of recent delays.[2] However, these assumptions are tentative as the project's feasibility

[2] As before, construction under the water component is expected to begin in 2014-15 and would run through 2021/22. The total cost of this component (US$900 million) will be financed by grants from South Africa.

study is still underway (expected to be completed by end-2014). The actual size of the project could be larger and the terms of the financing could differ.

4. In line with the current policy, this update also uses a discount rate of 5 percent.

EXTERNAL DEBT SUSTAINABILITY

A. Baseline

5. All external debt sustainability indicators remain below their corresponding thresholds in the baseline scenario (Table 1a). For example, the present value (PV) of external debt to GDP ratio is expected to fall to below 30 percent in 2016, after which time it is expected to rise to 39 percent in 2022, before subsiding to 34 percent in the early 2030s. The temporary rise in the debt ratio is driven by the construction of a hydropower plant under LHWP-2 and related financing needs. Moreover, in the long run the looser fiscal outlook (higher primary deficits) raises the PV of external debt to GDP ratio by more than 10 percent of GDP, compared with the previous DSA.

6. These results are sensitive to the assumptions concerning the amount and concessionality of LHWP-2 financing, the future of SACU revenues, and the concessionality of external borrowing in general. The temporary increase in PV of external debt to GDP ratio is driven entirely by higher nonconcessional borrowing for the LHWP-2. The increased borrowing during the construction of the power plant does make Lesotho temporarily more vulnerable to external shocks. The authorities are currently undertaking a feasibility study, results of which would provide a better idea about the financing options. While the baseline assumes that the government revenue would gradually decline from 56 percent of GDP in 2014 to 48 percent of GDP in 2023 and onwards, largely due to a decline in SACU revenue, a more abrupt and permanent decline of these revenues—as discussed in the risk assessment matrix in the staff report for Lesotho's Article IV consultation (EBS/14/[ ])—would require a major fiscal adjustment to maintain macroeconomic stability rather than additional borrowing.

B. Sensitivity Analysis

7. Stress tests show that Lesotho's external debt vulnerabilities would increase significantly, in particular in the event of a worsening of borrowing terms, a major exchange rate depreciation, or if the export growth turned out to be lower than the historical average. (Table 1b). A 200 basis point increase in interest rates for new public sector loans in 2013–33 (A2) would increase the PV of external debt-to-GDP ratio to 55 percent in 2024 and beyond. Similarly, in the event of a one-time 30 percent depreciation of the nominal exchange rate (B6) in 2014, the PV of external debt-to-GDP ratio would increase to 55 percent by 2021/22, gradually falling thereafter to 50 percent by 2028 and staying at about 48 percent thereafter. In both of these stress tests, the 40-percent threshold would be breached throughout the projection period. In a scenario where export value grows (B2) at one standard deviation lower than the historical average, the PV of external debt-to-export ratio would rise from 2018 and peak in 2021 at 142 percent, before easing to 100 percent in 2030s. In a scenario in which the key variables are set at their average of the past 10 years, Lesotho's external debt ratios actually fall relative to the baseline, reflecting an average fiscal surplus over the past years thanks to the large SACU revenue in 2006–09. However, given the structural break, the historical scenario could be considered less relevant for the analysis.

C. Including Remittances[3]

8. When remittances are included in the analysis, a key external debt indicator would be briefly breached, albeit by a small margin. (Figure 2). The PV of external debt to GDP plus remittances ratio is expected to fall to 28 percent by 2017, after which time it is expected to rise, reaching nearly 36½ percent in 2021–23 (the relevant threshold is 36 percent), and then fall to 32 percent by the end of the projection period. Other indicators behave similarly to the analysis excluding remittances.

PUBLIC SECTOR DEBT SUSTAINABILITY

9. Public debt indicators largely mirror those of external debt, because domestic debt remains low throughout the projection period (2–4 percent of GDP). Domestic debt is projected to fall to about 2 percent of GDP by 2018/19, and thereafter to gradually increase to about 4 percent of GDP in 2032/33. The PV of public sector debt stood at 37 percent in 2012/13.

10. The standard sensitivity tests for public debt distress point to higher degree of vulnerability of public debt to lower long-run GDP growth (Table 2b). In the case of permanently lower GDP growth (A3) the PV of debt to GDP ratio would increase permanently throughout the projection period to 56 percent. The permanently-lower-GDP-growth test has a similar effect on the PV of debt-to-revenue and debt service-to-revenue ratios.

CONCLUSION

11. Lesotho remains at moderate risk of external debt distress. All external debt and debt service indicators for the base case remain below their respective thresholds. However, there is a permanent breach in the sensitivity tests. Compared with the previous DSA, the primary fiscal deficit has deteriorated throughout the projection period by an average of about 4 percent of GDP a year, increasing the PV of external debt to GDP ratio by 12 percentage points at the end of the projection period. The risks appear manageable, provided that the fiscal stance does not deteriorate further and that government's external borrowing is largely on concessional terms. Uncertainty about the parameters of the LHWP-2 project and its financing could also create some risk to debt sustainability. For the next DSA, these parameters should be updated based on the conclusions of the project's feasibility study currently underway.

12. The authorities broadly agreed with the assessment of moderate risk of external debt distress. They appreciated that the PV of external debt ratio rises owing to the construction of the LHWP-2 and agreed that care would be needed to ensure that the financing arrangements do not increase the risk of external debt distress.

[3] Presenting the analysis including remittances is mainly for illustrative purposes. With declining migrant employment in South Africa's mining sector, Lesotho's dependence on remittances continues to diminish. In the long term, the ratio of remittances to GDP is expected to fall well below 10 percent.

Table 1a .Lesotho: External Debt Sustainability Framework, Baseline Scenario, 2011–33 [1]

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections								
	2011	2012	2013			2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
External debt (nominal) 1/	**31.5**	**36.5**	**38.9**			**37.8**	**38.3**	**39.1**	**41.0**	**43.8**		**53.2**	**49.4**	
of which: public and publicly guaranteed (PPG)	31.5	36.5	38.9			37.8	38.3	39.1	41.0	43.8		53.2	49.4	
Change in external debt	31.5	5.0	2.5			-1.1	0.5	0.8	1.9	2.8		0.2	-0.4	
Identified net debt-creating flows	...	4.3	-1.2			-0.7	7.0	8.3	6.2	7.7		4.1	-1.4	
Non-interest current account deficit	**8.6**	**2.4**	**0.6**	**-9.3**	**13.5**	**0.4**	**8.2**	**9.5**	**7.7**	**9.3**		**5.9**	**2.2**	4.7
Deficit in balance of goods and services	61.1	66.0	62.5			60.1	63.9	61.4	58.3	58.5		53.6	49.4	
Exports	46.3	44.5	41.4			41.7	40.3	42.3	43.2	43.1		41.8	45.0	
Imports	107.4	110.5	103.9			101.8	104.2	103.7	101.5	101.6		95.3	94.4	
Net current transfers (negative = inflow)	-27.2	-40.7	-38.7	-38.6	7.1	-38.3	-35.8	-32.7	-33.2	-32.6		-32.5	-34.4	-33.2
of which: official	-20.0	-33.8	-31.7			-32.1	-30.1	-26.8	-27.2	-26.7		-26.5	-27.9	
Other current account flows (negative = net inflow)	-25.3	-22.9	-23.2			-21.4	-19.9	-19.3	-17.5	-16.5		-15.1	-12.8	
Net FDI (negative = inflow)	**-0.4**	**-0.4**	**-0.3**	**-2.2**	**1.6**	**-0.3**	**-0.3**	**-0.3**	**-0.4**	**-0.4**		**-0.7**	**-2.0**	-1.1
Endogenous debt dynamics 2/	**...**	**2.3**	**-1.5**			**-0.8**	**-0.9**	**-0.8**	**-1.1**	**-1.2**		**-1.1**	**-1.6**	
Contribution from nominal interest rate	...	0.4	0.5			0.6	0.8	1.0	0.9	0.9		1.1	0.5	
Contribution from real GDP growth	0.0	-2.2	-2.0			-1.4	-1.7	-1.8	-2.0	-2.1		-2.2	-2.1	
Contribution from price and exchange rate changes	0.0	4.1	...			...	...	...	...	...		...	...	
Residual (3-4) 3/	**...**	**0.7**	**1.8**			**0.7**	**-5.3**	**-6.8**	**-3.7**	**-4.4**		**-2.8**	**2.0**	
of which: exceptional financing	...	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	33.4	32.7			30.6	29.9	29.6	30.5	32.3		38.1	34.2	
In percent of exports	...	75.1	79.1			73.2	74.2	70.0	70.5	74.9		91.3	76.1	
PV of PPG external debt	**...**	**33.4**	**32.7**			**30.6**	**29.9**	**29.6**	**30.5**	**32.3**		**38.1**	**34.2**	
In percent of exports	**...**	**75.1**	**79.1**			**73.2**	**74.2**	**70.0**	**70.5**	**74.9**		**91.3**	**76.1**	
In percent of government revenues	**...**	**58.1**	**61.1**			**54.3**	**54.9**	**57.4**	**60.0**	**64.3**		**79.1**	**71.0**	
Debt service-to-exports ratio (in percent)	**3.3**	**3.2**	**4.2**			**4.2**	**5.5**	**5.0**	**4.6**	**4.8**		**9.0**	**5.6**	
PPG debt service-to-exports ratio (in percent)	**3.3**	**3.2**	**4.2**			**4.2**	**5.5**	**5.0**	**4.6**	**4.8**		**9.0**	**5.6**	
PPG debt service-to-revenue ratio (in percent)	**3.4**	**2.5**	**3.3**			**3.1**	**4.1**	**4.1**	**3.9**	**4.1**		**7.8**	**5.2**	
Total gross financing need (Billions of U.S. dollars)	241	80	47			47	272	323	287	361		413	237	
Non-interest current account deficit that stabilizes debt ratio	-22.8	-2.6	-1.9			1.6	7.7	8.6	5.8	6.6		5.7	2.6	
Key macroeconomic assumptions														
Real GDP growth (in percent)	3.8	6.7	5.4	4.5	1.3	4.0	4.9	5.0	5.5	5.5	4.5	4.5	4.5	4.7
GDP deflator in US dollar terms (change in percent)	3.7	-11.6	-4.9	9.7	18.3	2.9	3.2	1.8	1.5	1.2	4.2	2.0	2.0	1.9
Effective interest rate (percent) 5/	...	1.2	1.3	1.2	...	1.8	2.3	2.7	2.4	2.4	2.0	2.3	1.1	1.7
Growth of exports of G&S (US dollar terms, in percent)	15.3	-9.2	-6.9	10.1	15.5	7.8	4.5	12.1	9.6	6.5	7.4	5.6	7.0	7.0
Growth of imports of G&S (US dollar terms, in percent)	6.5	-2.9	-5.9	11.1	15.1	4.9	10.8	6.3	4.9	6.8	6.7	2.7	7.0	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	40.4	34.0	33.7	28.9	26.1	32.6	38.5	42.3	37.7
Government revenues (excluding grants, in percent of GDP)	44.6	57.5	53.6			56.2	54.4	51.5	50.8	50.2		48.2	48.2	48.3
Aid flows (in Billions of US dollars) 7/	893	942	928			855	824	800	780	758		594	368	520
of which: Grants	193	199	124			105	102	104	113	121		165	312	207
of which: Concessional loans	700	744	804			750	721	695	668	637		429	56	313
Grant-equivalent financing (in percent of GDP) 8/	...	...	5.3			6.5	5.9	5.5	5.6	5.7		5.9	5.6	5.6
Grant-equivalent financing (in percent of external financing)	...	...	100.0			66.0	59.1	59.8	53.8	50.0		61.1	67.2	62.2
Memorandum items:														
Nominal GDP (Billions of US dollars)	2,461	2,321	2,325			2,486	2,692	2,877	3,081	3,289		4,620	8,747	
Nominal dollar GDP growth	7.6	-5.7	0.2			6.9	8.3	6.9	7.1	6.8	6.0	6.6	6.6	6.7
PV of PPG external debt (in Billions of US dollars)		731	721			760	804	851	939	1,062		1,762	2,993	
(PVt-PVt-1)/GDPt-1 (in percent)			-0.4			1.7	1.8	1.7	3.1	4.0	2.0	1.9	2.0	2.6
Gross workers' remittances (Billions of US dollars)	298	260	240			227	203	185	170	160		283	583	
PV of PPG external debt (in percent of GDP + remittances)	...	30.1	29.7			28.0	27.8	27.8	28.9	30.8		35.9	32.1	
PV of PPG external debt (in percent of exports + remittance	...	60.0	63.3			60.1	62.5	60.8	62.6	67.3		79.6	66.3	
Debt service of PPG external debt (in percent of exports + r	...	2.5	3.4			3.5	4.7	4.3	4.1	4.3		7.9	4.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–33

(In percent)

	Projections						
	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio							
Baseline	31	30	30	30	32	**38**	34
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	21	7	-7	-17	-27	**-63**	-56
A2. New public sector loans on less favorable terms in 2013-2033 2	30	32	33	36	40	**55**	54
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	31	31	30	31	33	**39**	35
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	34	39	38	39	40	**45**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	34	38	38	39	41	**48**	43
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	35	37	36	37	39	**43**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	35	39	38	39	41	**46**	37
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	43	42	42	43	46	**54**	48
PV of debt-to-exports ratio							
Baseline	73	74	70	71	75	**91**	76
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	50	16	-17	-40	-62	**-150**	-123
A2. New public sector loans on less favorable terms in 2013-2033 2	73	79	79	84	93	**132**	121
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	73	74	70	70	75	**91**	76
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	93	121	114	113	118	**136**	97
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	73	74	70	70	75	**91**	76
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	84	91	86	86	90	**104**	77
B5. Combination of B1-B4 using one-half standard deviation shocks	86	97	91	91	96	**111**	83
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	73	74	70	70	75	**91**	76
PV of debt-to-revenue ratio							
Baseline	54	55	57	60	64	**79**	71
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	37	12	-14	-34	-53	**-130**	-115
A2. New public sector loans on less favorable terms in 2013-2033 2	54	59	65	71	80	**115**	112
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	55	56	59	61	66	**81**	73
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	60	71	74	77	81	**94**	72
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	61	70	73	76	82	**101**	90
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	62	68	70	73	77	**90**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	62	71	74	76	81	**95**	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	77	78	81	85	91	**112**	100

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 1b. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-33 (concluded)

(In percent)

Debt service-to-exports ratio							
Baseline	4	6	5	5	5	**9**	6
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	4	4	3	2	2	**0**	-8
A2. New public sector loans on less favorable terms in 2013-2033 2	4	6	5	5	5	**9**	10
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	6	5	5	5	**9**	6
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	7	7	6	6	**13**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	6	5	5	5	**9**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	6	5	5	5	**10**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	6	6	5	5	**11**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	6	5	5	5	**9**	6
Debt service-to-revenue ratio							
Baseline	3	4	4	4	4	**8**	5
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	3	3	3	2	1	**0**	-8
A2. New public sector loans on less favorable terms in 2013-2033 2	3	4	4	4	5	**8**	9
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	3	4	4	4	4	**8**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	3	4	4	4	4	**9**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	5	5	5	5	**10**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	3	4	4	4	4	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	5	4	5	**9**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	6	6	6	6	**11**	7
Memorandum item:							
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	**33**	33

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Lesotho: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–33

(In percent of GDP, unless otherwise indicated)

	Actual					Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	37.6	39.7	41.8			40.4	40.6	41.3	43.0	45.8		56.3	53.4	
of which: foreign-currency denominated	31.5	36.5	38.9			37.8	38.3	39.1	41.0	43.8		53.2	49.4	
							2.4	2.1	1.9	1.9		3.0	4.0	
Change in public sector debt	32.8	2.1	2.1			-1.4	0.2	0.6	1.7	2.8		0.3	-0.3	
Identified debt-creating flows	...	-2.5	0.6			-2.9	-1.9	0.4	0.8	1.0		2.1	1.7	
Primary deficit	9.9	-5.8	0.6	-3.4	7.5	1.0	0.3	2.0	2.6	2.8	1.7	4.3	4.3	4.2
Revenue and grants	52.5	66.1	58.9			60.5	58.2	55.2	54.5	53.9		51.8	51.8	51.9
of which: grants	7.8	8.6	5.3			4.2	3.8	3.6	3.7	3.7		3.6	3.6	
Primary (noninterest) expenditure	62.4	60.3	59.5			61.5	58.4	57.1	57.1	56.7		56.1	56.1	56.1
Automatic debt dynamics	...	3.3	0.0			-3.9	-2.2	-1.5	-1.8	-1.7		-2.2	-2.6	
Contribution from interest rate/growth differential	...	-2.1	-2.1			-1.4	-1.6	-1.5	-1.9	-2.1		-2.2	-2.6	
of which: contribution from average real interest rat	...	0.3	-0.1			0.2	0.3	0.4	0.2	0.2		0.2	-0.3	
of which: contribution from real GDP growth	-0.2	-2.4	-2.0			-1.6	-1.9	-1.9	-2.1	-2.2		-2.4	-2.3	
Contribution from real exchange rate depreciation	...	5.3	2.1			-2.5	-0.6	-0.1	0.2	0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	...	4.7	1.5			1.5	2.1	0.2	0.9	1.8		-1.8	-2.0	
Other Sustainability Indicators														
PV of public sector debt	...	36.7	35.6			33.2	32.2	31.7	32.4	34.2		41.2	38.2	
of which: foreign-currency denominated	...	33.4	32.7			30.6	29.9	29.6	30.5	32.3		38.1	34.2	
of which: external	...	33.4	32.7			30.6	29.9	29.6	30.5	32.3		38.1	34.2	
PV of contingent liabilities (not included in public sectc	...	...	...			...	...	...	...	...		...	...	
Gross financing need 2/	12.5	-3.6	2.9			3.2	2.9	4.4	4.9	5.1		8.7	7.6	
PV of public sector debt-to-revenue and grants ratio (in perc	...	55.5	60.5			54.9	55.4	57.5	59.5	63.5		79.5	73.9	
PV of public sector debt-to-revenue ratio (in percent)	...	63.7	66.5			59.0	59.3	61.5	63.8	68.1		85.4	79.3	
of which: external 3/	...	58.1	61.1			54.3	54.9	57.4	60.0	64.3		79.1	71.0	
Debt service-to-revenue and grants ratio (in percent) 4/	4.9	3.2	3.9			3.7	4.6	4.4	4.3	4.4		8.5	6.3	
Debt service-to-revenue ratio (in percent) 4/	5.8	3.7	4.3			3.9	4.9	4.8	4.6	4.7		9.2	6.8	
Primary deficit that stabilizes the debt-to-GDP ratio	-22.9	-7.9	-1.5			2.4	0.1	1.4	0.9	0.0		4.0	4.6	
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	3.8	6.7	5.4	4.5	1.3	4.0	4.9	5.0	5.5	5.5	4.5	4.5	4.5	4.7
Average nominal interest rate on forex debt (in percent)	...	1.2	1.3	1.2	...	1.8	2.3	2.7	2.4	2.4	2.0	2.3	1.1	1.7
Average real interest rate on domestic debt (in percent)	0.4	5.8	0.0	3.1	3.8	2.7	1.8	2.8	3.0	3.0	2.5	2.9	2.9	2.9
Real exchange rate depreciation (in percent, + indicates dep	7.5	18.1	6.1	-1.5	13.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.4	1.7	7.6	6.3	3.3	5.2	6.1	5.1	4.8	4.8	5.4	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, i	14.7	3.3	3.8	0.8	6.1	7.5	-0.3	2.7	5.3	4.8	3.8	4.5	4.5	4.6
Grant element of new external borrowing (in percent)	...	...	0.0	...	...	40.4	34.0	33.7	28.9	26.1	27.2	38.5	42.3	...

Sources: Country authorities; and staff estimates and projections.

1/ Gross debt is used. The public sector comprises the central government, the Central Bank of Lesotho, and all enterprises with majority state ownership.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Lesotho: Sensitivity Analysis for Key Indicators of Public Debt 2014–33

(In percent)

	Projections						
	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio							
Baseline	33	32	32	32	34	41	38
A. Alternative scenarios							
A1. Real GDP growth and primary balance are at historical averages	30	27	23	20	19	5	-33
A2. Primary balance is unchanged from 2013	33	32	31	30	31	27	6
A3. Permanently lower GDP growth 1/	33	33	33	34	36	47	56
B. Bound tests							
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	34	34	34	36	38	49	51
B2. Primary balance is at historical average minus one standard deviations in 2014-201	35	37	36	37	38	45	40
B3. Combination of B1-B2 using one half standard deviation shocks	33	32	32	33	35	44	44
B4. One-time 30 percent real depreciation in 2014	45	42	41	40	41	46	43
B5. 10 percent of GDP increase in other debt-creating flows in 2014	40	39	38	38	40	46	41
PV of Debt-to-Revenue Ratio 2/							
Baseline	55	55	58	60	63	80	74
A. Alternative scenarios							
A1. Real GDP growth and primary balance are at historical averages	50	46	42	37	35	9	-64
A2. Primary balance is unchanged from 2013	54	55	56	55	57	53	12
A3. Permanently lower GDP growth 1/	55	56	59	62	67	90	108
B. Bound tests							
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	56	59	62	66	71	94	97
B2. Primary balance is at historical average minus one standard deviations in 2014-201	58	63	66	67	71	87	78
B3. Combination of B1-B2 using one half standard deviation shocks	54	55	58	61	66	85	84
B4. One-time 30 percent real depreciation in 2014	74	73	74	74	77	88	83
B5. 10 percent of GDP increase in other debt-creating flows in 2014	66	67	69	71	74	89	79
Debt Service-to-Revenue Ratio 2/							
Baseline	4	5	4	4	4	9	6
A. Alternative scenarios							
A1. Real GDP growth and primary balance are at historical averages	4	4	4	4	3	7	-3
A2. Primary balance is unchanged from 2013	4	5	4	4	4	8	3
A3. Permanently lower GDP growth 1/	4	5	5	4	4	9	8
B. Bound tests							
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	4	5	5	5	5	9	8
B2. Primary balance is at historical average minus one standard deviations in 2014-201	4	5	5	5	5	9	7
B3. Combination of B1-B2 using one half standard deviation shocks	4	5	4	4	4	9	7
B4. One-time 30 percent real depreciation in 2014	4	6	6	6	6	12	10
B5. 10 percent of GDP increase in other debt-creating flows in 2014	4	5	6	5	5	9	7

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 3a. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt (including remittances), 2014-33

(In percent)

	Projections						
	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to-GDP+remittances ratio							
Baseline	28	28	28	29	31	**36**	32
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	19	6	-7	-16	-26	**-61**	-55
A2. New public sector loans on less favorable terms in 2013-2033 2	28	30	31	34	38	**52**	51
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	28	28	28	30	31	**37**	33
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	31	36	36	37	39	**42**	33
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	31	35	35	36	39	**45**	40
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	32	34	34	35	37	**41**	32
B5. Combination of B1-B4 using one-half standard deviation shocks	32	36	36	37	39	**43**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	38	38	38	40	43	**50**	44
PV of debt-to-exports+remittances ratio							
Baseline	60	62	61	63	67	**80**	66
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	41	14	-15	-36	-57	**-139**	-119
A2. New public sector loans on less favorable terms in 2013-2033 2	60	67	69	74	84	**115**	105
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	60	62	61	63	67	**80**	66
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	74	98	96	98	104	**115**	82
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	60	62	61	63	67	**80**	66
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	71	78	75	76	81	**91**	67
B5. Combination of B1-B4 using one-half standard deviation shocks	71	81	78	80	85	**95**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	60	62	61	63	67	**80**	66
PV of debt-to-revenue ratio							
Baseline	54	55	57	60	64	**79**	71
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	37	12	-14	-34	-53	**-130**	-115
A2. New public sector loans on less favorable terms in 2013-2033 2	54	59	65	71	80	**115**	112
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	55	56	59	61	66	**81**	73
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	60	71	74	77	81	**94**	72
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	61	70	73	76	82	**101**	90
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	62	68	70	73	77	**90**	72
B5. Combination of B1-B4 using one-half standard deviation shocks	62	71	74	76	81	**95**	76
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	77	78	81	85	91	**112**	100

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3a. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt (including remittances), 2014–33 (concluded)

(In percent)

Debt service-to-exports+remittances ratio							
Baseline	3	5	4	4	4	**8**	5
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	3	4	3	2	1	**0**	-8
A2. New public sector loans on less favorable terms in 2013-2033 2	3	5	5	5	5	**8**	9
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	3	5	4	4	4	**8**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	6	6	5	6	**11**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	3	5	4	4	4	**8**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	5	5	4	5	**9**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	5	5	5	**9**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	3	5	4	4	4	**8**	5
Debt service-to-revenue ratio							
Baseline	3	4	4	4	4	**8**	5
A. Alternative Scenarios							
A1. Key variables at their historical averages in 2013-2033 1/	3	3	3	2	1	**0**	-8
A2. New public sector loans on less favorable terms in 2013-2033 2	3	4	4	4	5	**8**	9
B. Bound Tests							
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	3	4	4	4	4	**8**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	3	4	4	4	4	**9**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	5	5	5	5	**10**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	3	4	4	4	4	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	5	4	5	**9**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	6	6	6	6	**11**	7
Memorandum item:							
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	**33**	33

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

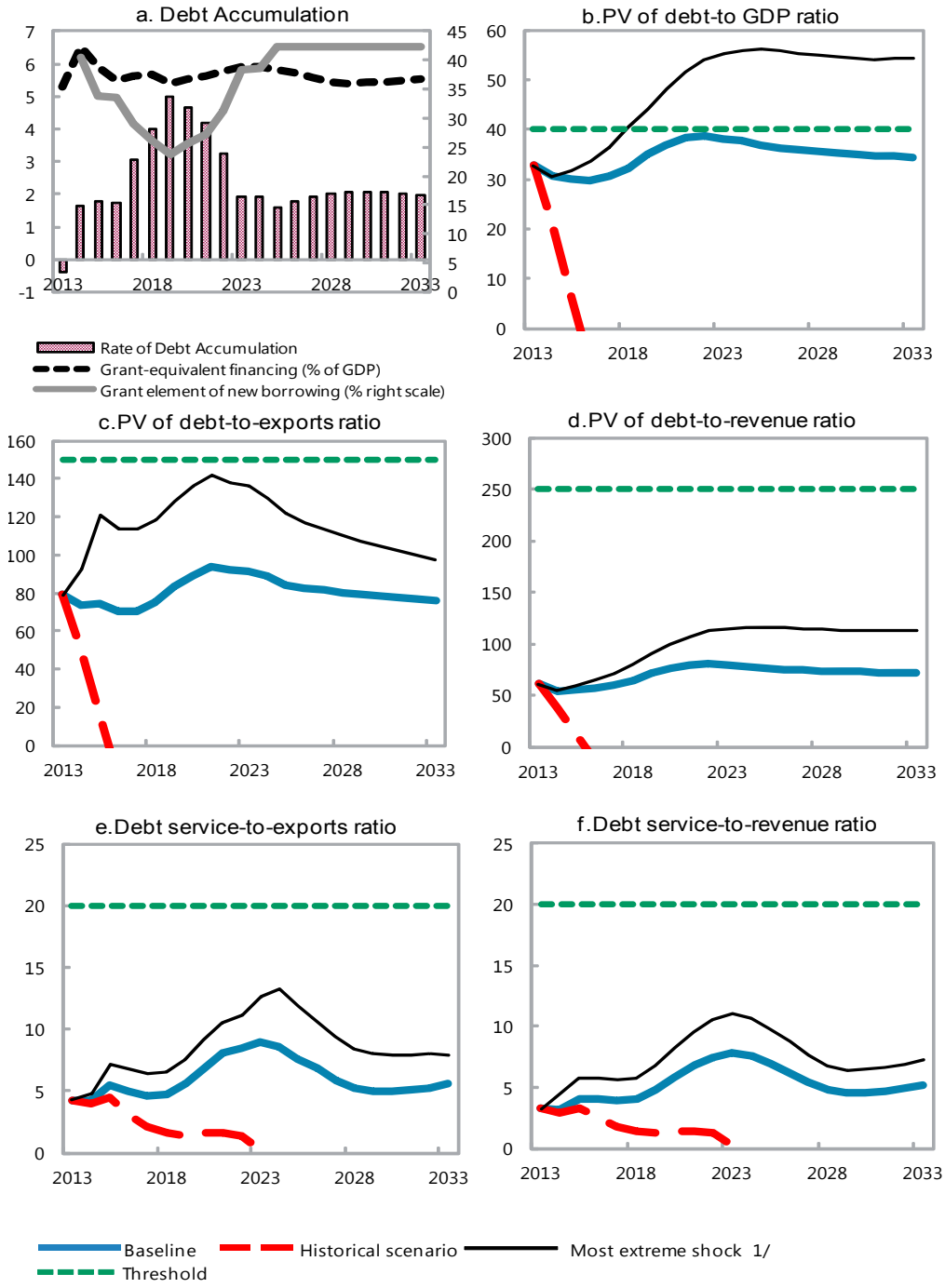
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

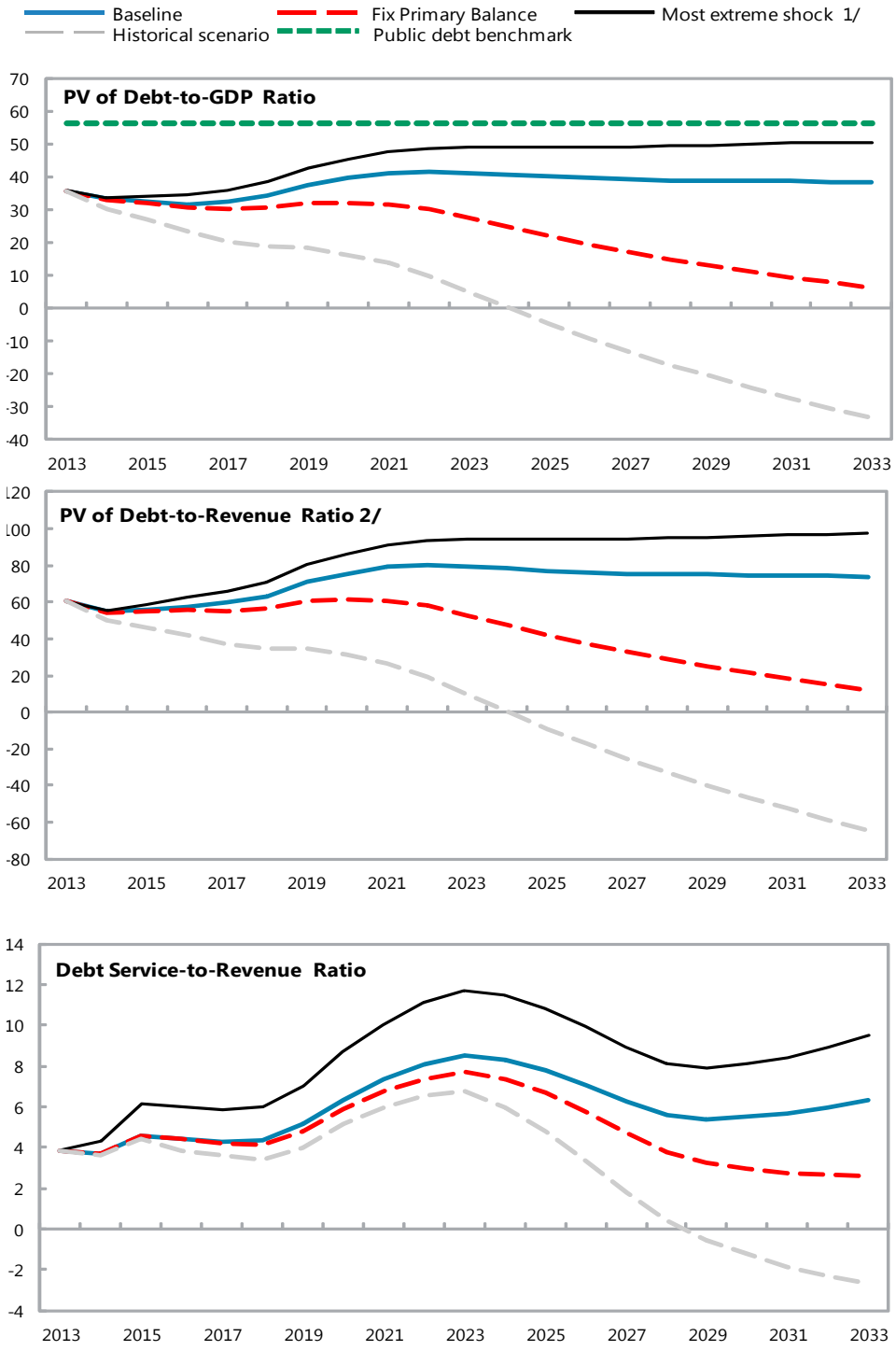
Figure 1a. Lesotho: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Country authorities; and staff estimates and projections.

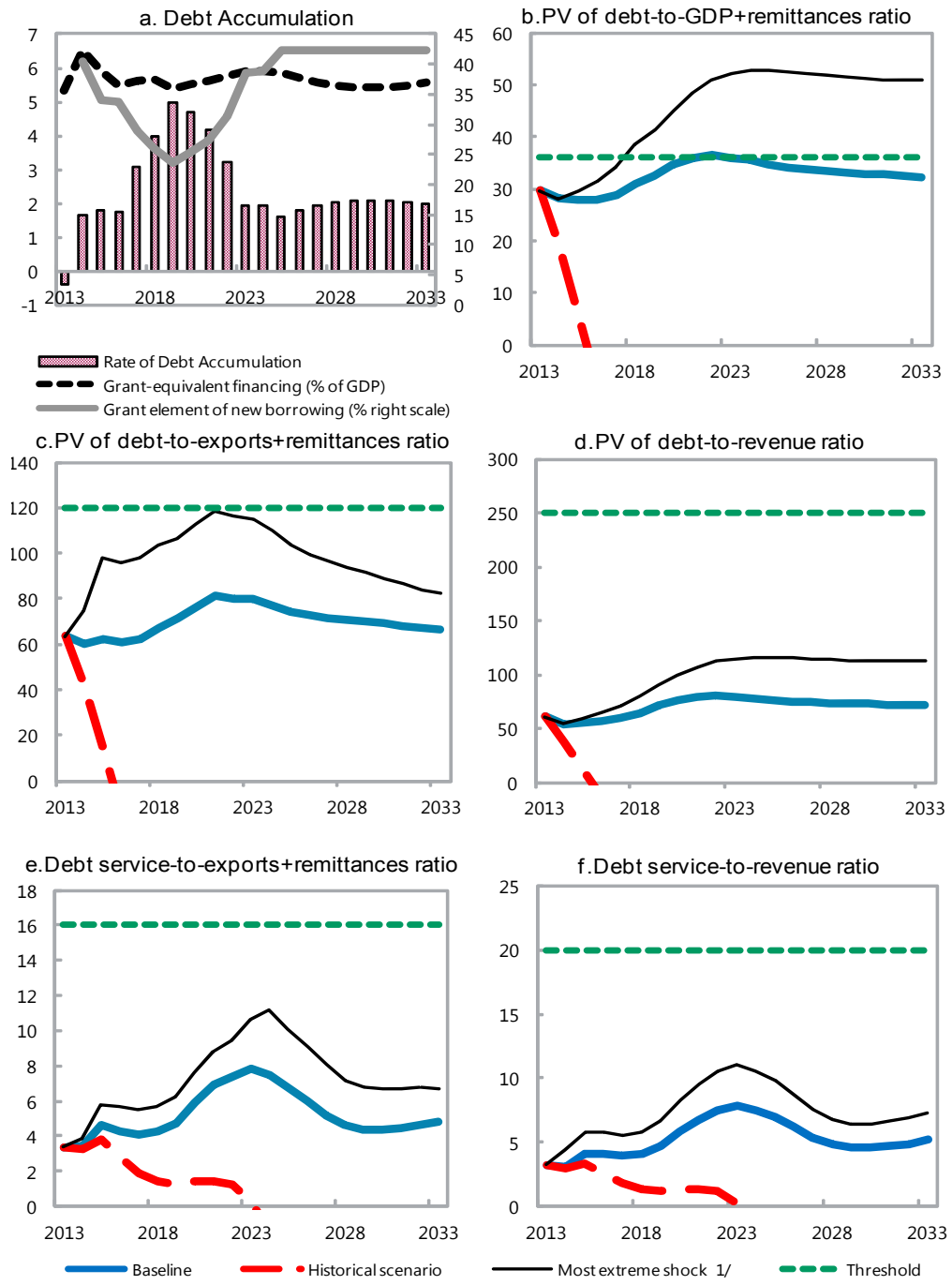
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a Terms shock; in c. to a Exports shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Figure 1b.Lesotho: Indicators of Public Debt Under Alternative Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Figure 2. Lesotho: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios (including remittances), 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a Terms shock; in c. to a Exports shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.